```
!pip install chromadb
!pip install sentence-transformers
!pip install langchain
```

```
Collecting chromadb
  Downloading chromadb-1.5.6-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_
Collecting build>=1.0.3 (from chromadb)
  Downloading build-1.4.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: pydantic>=2.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: pydantic-settings>=2.0 in /usr/local/lib/python
Collecting pybase64>=1.4.1 (from chromadb)
  Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_6
Requirement already satisfied: uvicorn>=0.18.3 in /usr/local/lib/python3.12/di
Requirement already satisfied: numpy>=1.22.5 in /usr/local/lib/python3.12/dist
Requirement already satisfied: typing-extensions>=4.5.0 in /usr/local/lib/pyth
Collecting onnxruntime>=1.14.1 (from chromadb)
  Downloading onnxruntime-1.24.4-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_
Requirement already satisfied: opentelemetry-api>=1.2.0 in /usr/local/lib/pyth
Collecting opentelemetry-exporter-otlp-proto-grpc>=1.2.0 (from chromadb)
  Downloading opentelemetry_exporter_otlp_proto_grpc-1.40.0-py3-none-any.whl.m
Requirement already satisfied: opentelemetry-sdk>=1.2.0 in /usr/local/lib/pyth
Requirement already satisfied: tokenizers>=0.13.2 in /usr/local/lib/python3.12
Collecting pypika>=0.48.9 (from chromadb)
  Downloading pypika-0.51.1-py2.py3-none-any.whl.metadata (51 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 52.0/52.0 kB 1.2 MB/s eta 0:00:0
Requirement already satisfied: tqdm>=4.65.0 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: overrides>=7.3.1 in /usr/local/lib/python3.12/d
Requirement already satisfied: importlib-resources in /usr/local/lib/python3.1
Requirement already satisfied: grpcio>=1.58.0 in /usr/local/lib/python3.12/dis
Collecting bcrypt>=4.0.1 (from chromadb)
  Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl.metadata (10 kB
Requirement already satisfied: typer>=0.9.0 in /usr/local/lib/python3.12/dist-
Collecting kubernetes>=28.1.0 (from chromadb)
  Downloading kubernetes-35.0.0-py2.py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: tenacity>=8.2.3 in /usr/local/lib/python3.12/di
Requirement already satisfied: pyyaml>=6.0.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: mmh3>=4.0.1 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: orjson>=3.9.12 in /usr/local/lib/python3.12/dis
Requirement already satisfied: httpx>=0.27.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: jsonschema>=4.19.0 in /usr/local/lib/python3.12
Requirement already satisfied: packaging>=24.0 in /usr/local/lib/python3.12/di
Collecting pyproject_hooks (from build>=1.0.3->chromadb)
  Downloading pyproject_hooks-1.2.0-py3-none-any.whl.metadata (1.3 kB)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/lo
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.1
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/di
```

```
Requirement already satisfied: six>=1.9.0 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python
Requirement already satisfied: websocket-client!=0.40.0,!=0.41.*,!=0.42.*,>=0.
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: requests-oauthlib in /usr/local/lib/python3.12/
Requirement already satisfied: urllib3!=2.6.0,>=1.24.2 in /usr/local/lib/pytho
Collecting durationpy>=0.7 (from kubernetes>=28.1.0->chromadb)
  Downloading durationpy-0.10-py3-none-any.whl metadata (340 bytes)
```

```
import chromadb
from sentence_transformers import SentenceTransformer
```

```
client = chromadb.Client()

collection = client.create_collection(name="rag_collection")
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: Userw
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access publ
  warnings.warn(
modules.json: 100%                                          349/349 [00:00<00:00, 23.4kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated req
config_sentence_transformers.json: 100%                     116/116 [00:00<00:00, 8.43kB/s]
README.md:       10.5k/? [00:00<00:00, 676kB/s]
sentence_bert_config.json: 100%                             53.0/53.0 [00:00<00:00, 1.66kB/s]
config.json: 100%                                           612/612 [00:00<00:00, 14.6kB/s]
model.safetensors: 100%                                     90.9M/90.9M [00:01<00:00, 131MB/s]
Loading weights: 100%  103/103 [00:00<00:00, 461.35it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; r
tokenizer_config.json: 100%                                 350/350 [00:00<00:00, 14.2kB/s]
vocab.txt:       232k/? [00:00<00:00, 8.48MB/s]
tokenizer.json:       466k/? [00:00<00:00, 17.2MB/s]
special_tokens_map.json: 100%                               112/112 [00:00<00:00, 8.06kB/s]
config.json: 100%                                           190/190 [00:00<00:00, 11.9kB/s]
```

```
documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI.",
    "Deep Learning uses neural networks.",
    "Natural Language Processing deals with text data.",
    "RAG combines retrieval and generation."
]

ids = ["doc1", "doc2", "doc3", "doc4", "doc5"]
```

```
embeddings = model.encode(documents)
```

```
collection.add(
    documents=documents,
    embeddings=embeddings,
    ids=ids
)
```

```
query = "What is AI?"

query_embedding = model.encode([query])

results = collection.query(
    query_embeddings=query_embedding,
    n_results=2
)

print(results)
```

```
{'ids': [['doc1', 'doc2']], 'embeddings': None, 'documents': [['Artificial Intel
```

```
for doc in results['documents'][0]:
    print(doc)
```

```
Artificial Intelligence is the simulation of human intelligence.
Machine Learning is a subset of AI.
```